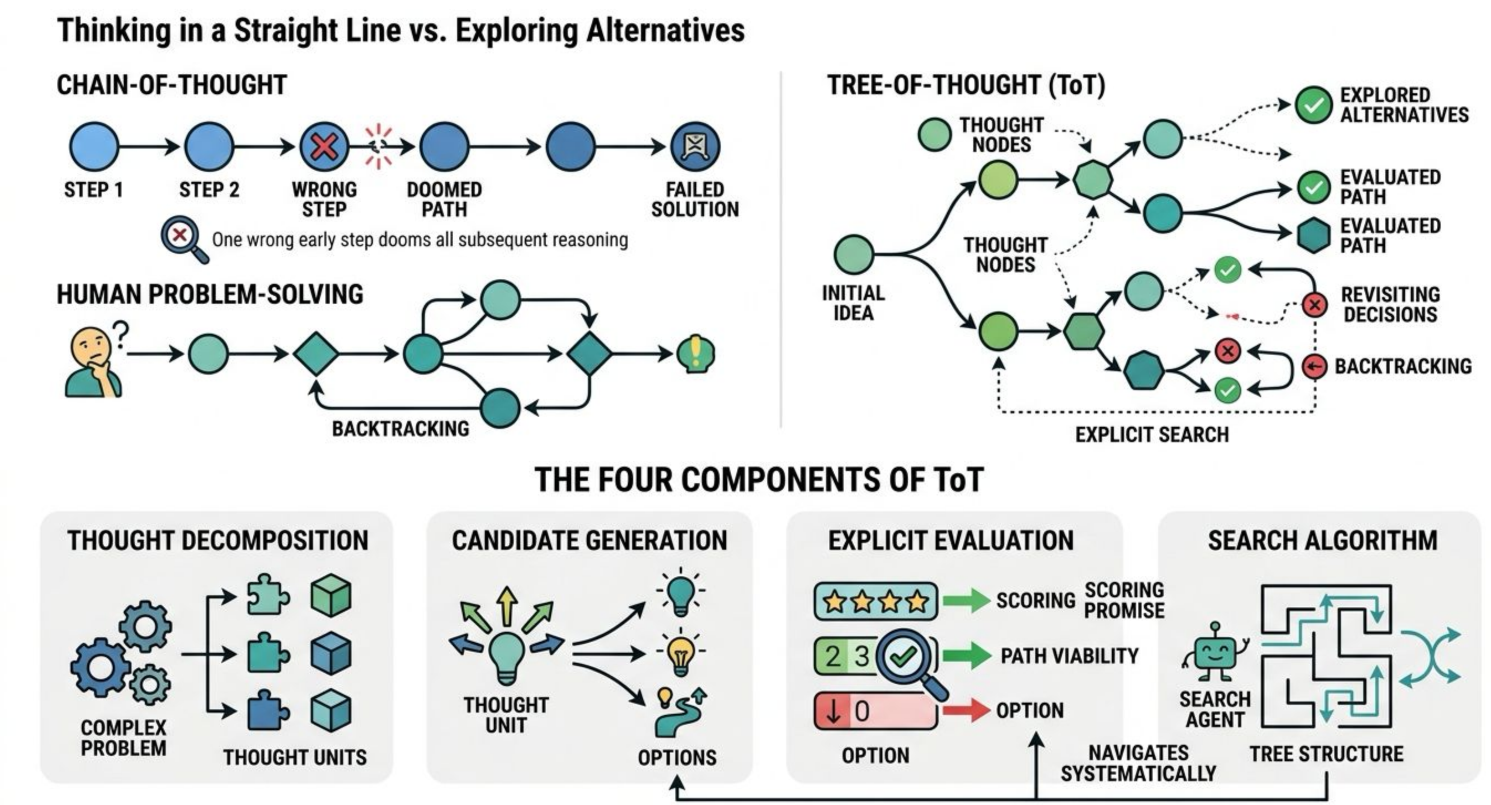
U.S. General Services Administration

Chapter 5.2: Tree-of-Thought (ToT) Fundamentals

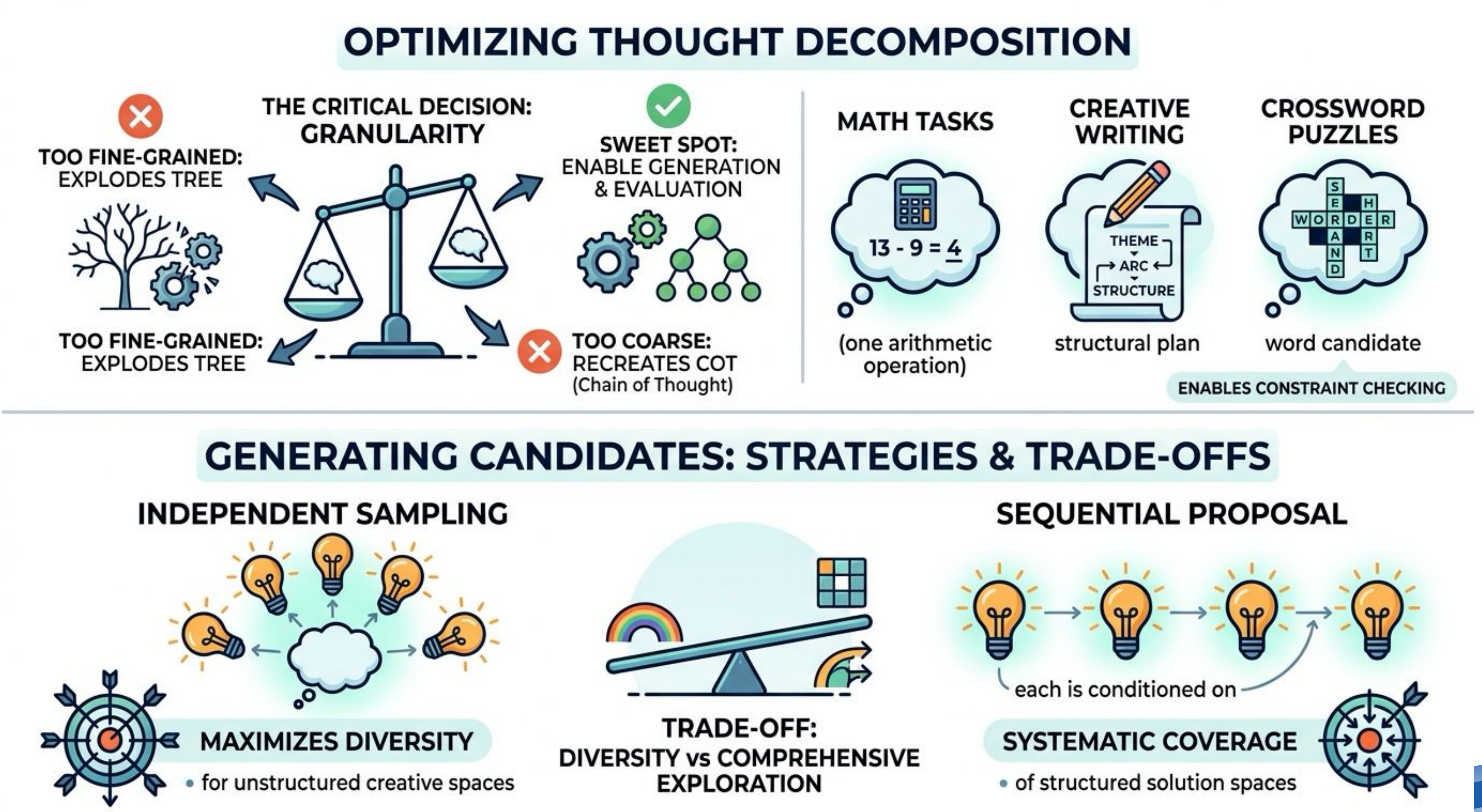
US General Service Administration

AI Community of Practice - March 2026

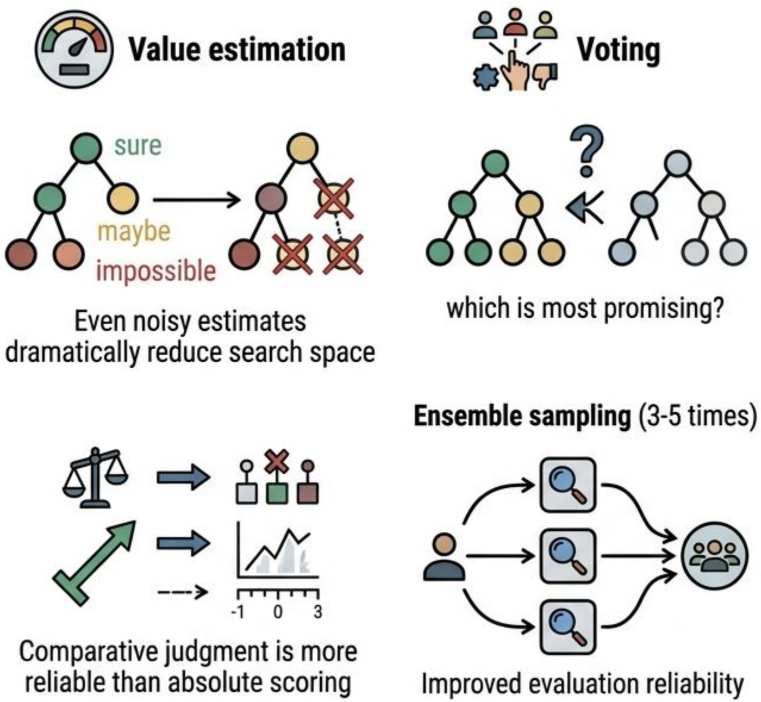
The Four Components of ToT



Thought Decomposition & Generating Candidates

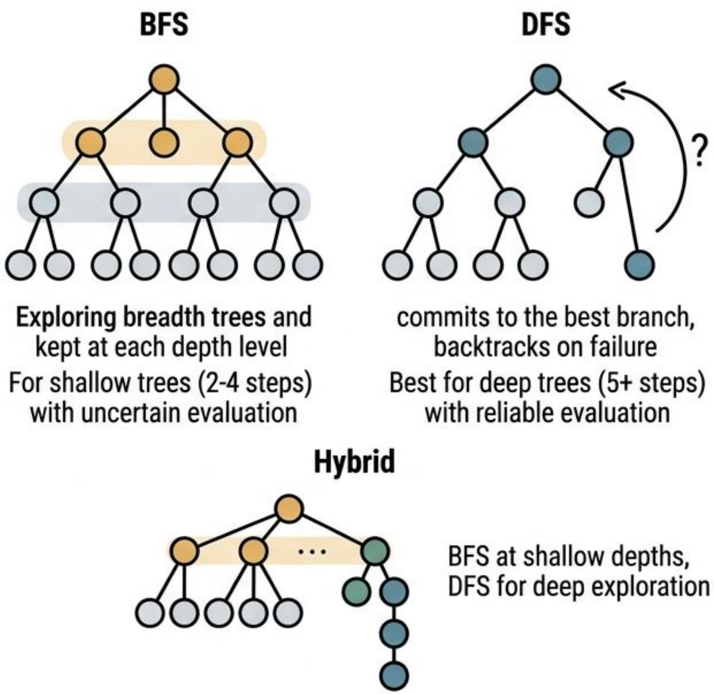


Evaluating Branches -- Value Scores and Voting



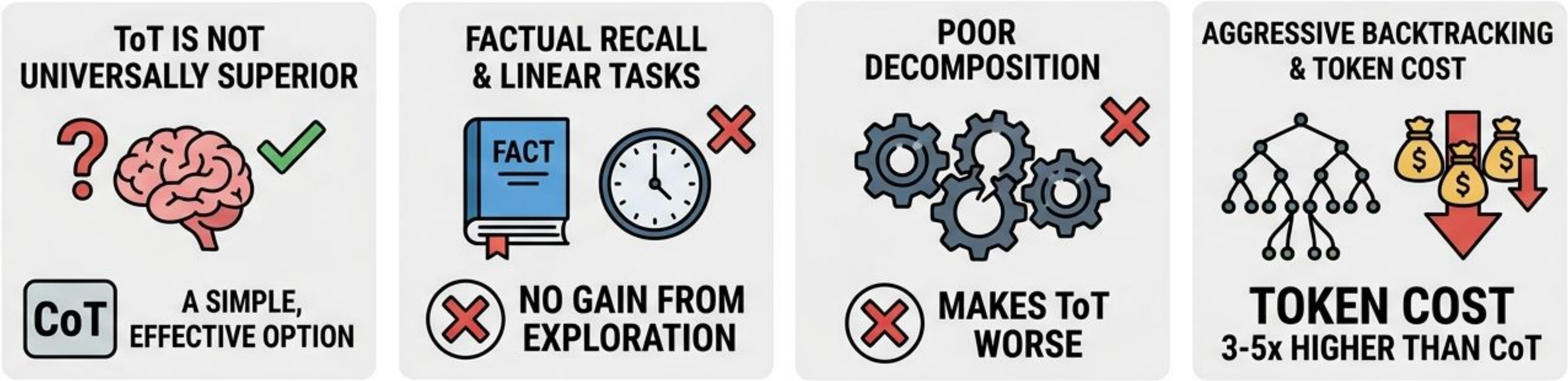
- Value estimation rates each state: "sure / maybe / impossible"
- Even noisy estimates dramatically reduce search space
- Voting compares candidates: "which is most promising?"
- Comparative judgment is more reliable than absolute scoring
- Ensemble sampling (3-5 times) improves evaluation reliability

Choosing a Search Strategy

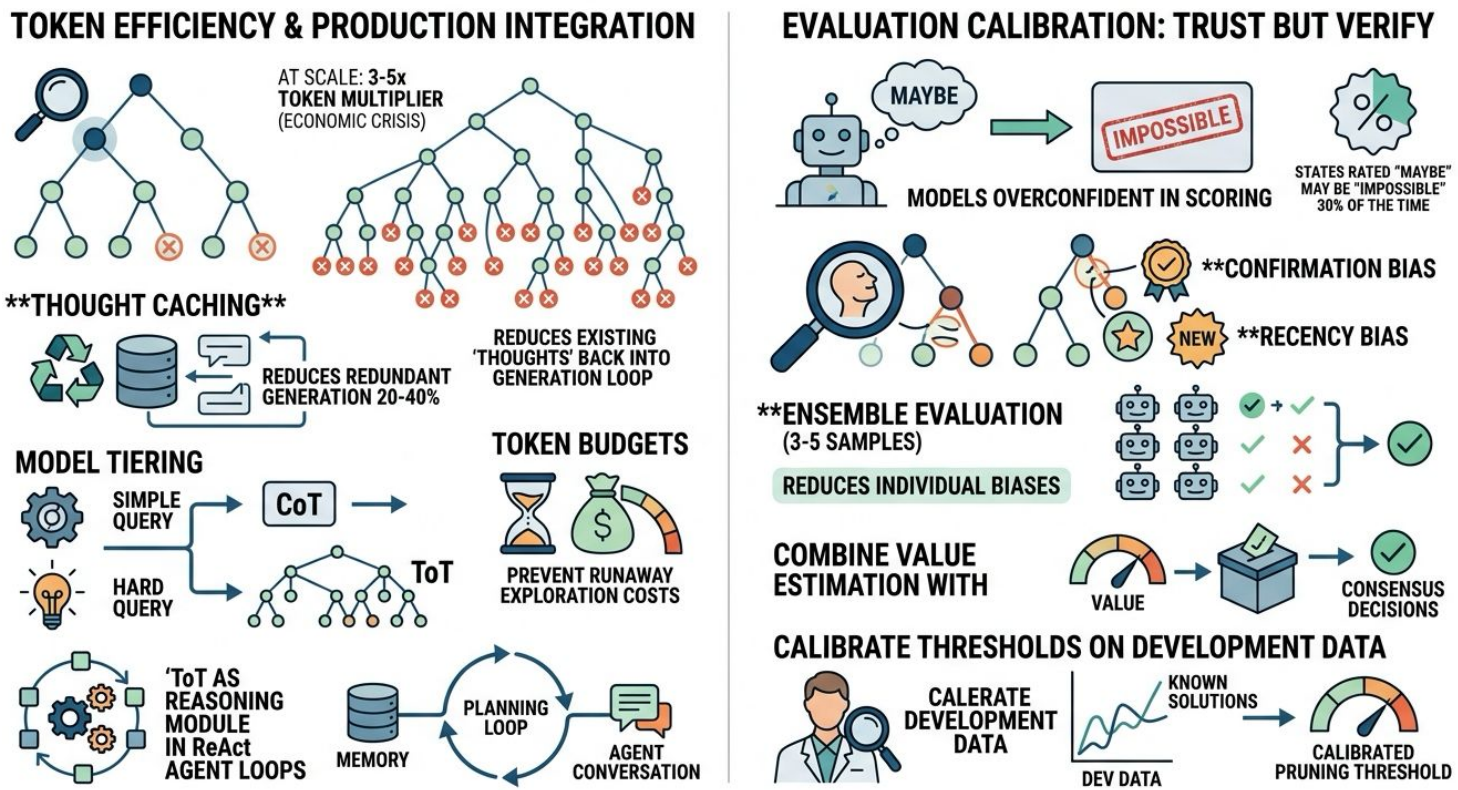


- BFS keeps top-b states at each depth level
- Best for shallow trees (2-4 steps) with uncertain evaluation
- DFS commits to the best branch, backtracks on failure
- Best for deep trees (5+ steps) with reliable evaluation
- Hybrid: BFS at shallow depths, DFS for deep exploration

When NOT to Use ToT



Token Efficiency and Evaluation Calibration



Real-World Applications of ToT